

DECISIONPRINT

Product Requirements Document

Living organizational decision memory powered by Hindsight

Core thesis: DecisionPrint remembers WHAT an organization decided, WHY it decided it, under WHICH constraints, WHAT happened afterward, and WHETHER those assumptions still hold.

Document	Value
Product	DecisionPrint
Primary platform dependency	Hindsight by Vectorize — central long-term memory engine
Target	AI-agent hackathon MVP
Judging emphasis	Innovation 30% · Hindsight Memory 25% · Technical 20% · UX 15% · Impact 10%
PRD version	1.0 · September 2026

This PRD is implementation-oriented. It deliberately separates Hindsight-native memory capabilities from deterministic application metadata and policy controls, so the hackathon build can be credible without recreating a second memory system.

Executive Summary

DecisionPrint is an AI-powered organizational decision-memory system. It turns fragmented project artifacts into a living record of decisions, their reasons, constraints, alternatives, assumptions, outcomes, exceptions, and later consequences. Its purpose is not document search. Its purpose is to reconstruct how organizational reasoning evolved over time and to make that history useful at the next decision point.

Hindsight is the product's memory engine, not an accessory. Source documents and transcripts are retained into an organizational memory bank. Hindsight extracts facts, temporal information, entities, and relationships; its recall operation retrieves through semantic, keyword, graph, and temporal strategies; background observations consolidate repeated evidence into evolving beliefs; reflect performs deeper reasoning over remembered material; and mental models can maintain standing organizational summaries. DecisionPrint adds the smallest deterministic layer required for canonical Decision objects, source permissions, comparison of old versus current constraints, drift flags, and auditability.

The hackathon MVP proves one loop exceptionally well: ingest historical project evidence; ingest a new project meeting; extract a decision and its context; retain that evidence in Hindsight; ask a new decision question; recall the relevant historical decision; compare historical and current constraints; surface Decision Drift; produce a source-backed decision brief; then ingest a later outcome or incident and show that the organizational memory has changed.

The demo must make one fact visually undeniable: without persistent longitudinal memory, “Kafka was rejected” looks like a rule; with Hindsight memory, it becomes “Kafka was rejected in 2025 because there were only two consumers and no replay requirement — conditions that no longer match this project.”

Problem Statement

Organizations document outputs better than reasoning. PRDs record requirements, architecture documents record chosen designs, tickets record work, meeting notes record fragments, and postmortems record failures. The connective tissue — why an option was chosen, which assumptions were true then, what alternatives were rejected, and what happened later — is scattered across time and tools.

This creates four recurring failures: institutional amnesia when people leave; cargo-cult reuse of old decisions after their constraints have changed; repeated exploration of already-tested alternatives; and weak organizational learning because downstream outcomes are not connected back to the decisions that preceded them.

Generic retrieval systems reduce search cost but do not solve the temporal reasoning problem. A similarity search can find “Do not use Kafka,” yet still miss that the statement was contingent on two consumers, moderate traffic, and no replay. DecisionPrint treats a decision as a time-bound claim embedded in a changing world.

Product Vision

Create an organizational memory that can answer not merely “What do our documents say?” but “What did we believe, why did we believe it, what evidence changed, and what should be reconsidered now?”

Product Thesis

The most valuable organizational knowledge is often conditional and longitudinal. A technical choice is meaningful only in relation to its constraints; an organizational principle is trustworthy only when its supporting evidence is visible; and a past decision should influence a new one only to the extent that the contexts are comparable.

Therefore, DecisionPrint treats persistent memory, temporal context, evidence consolidation, and evolving observations as first-class product primitives. Hindsight is uniquely central because the product's core behavior depends on retain → evolving memory → recall/reflect rather than one-shot retrieval.

Target Users

User	Primary need	DecisionPrint value
New employee	Understand why systems and practices exist	Evidence-backed organizational context, exceptions, mistakes, and historical rationale
Project / team lead	Make a new product or technical decision	Comparable prior decisions, old constraints, outcomes, exceptions, and drift
Executive / strategy user	See cross-project patterns without reading every project	Evidence-linked trends, recurring bottlenecks, capability demand, outdated assumptions

User Personas

Persona A — New Engineer: “Explain the organization to me”

Needs answers grounded in the company’s actual history, not generic model knowledge. Typical questions: Why do we use PostgreSQL here? Why was GraphQL rejected? What should I know before changing this subsystem? Which exceptions exist?

Persona B — Project Lead: “Help me avoid repeating history”

Needs a concise decision brief before committing to an architecture or product choice. The system must separate historical fact from inferred similarity and must explicitly compare old constraints with current constraints.

Persona C — CTO / COO: “Show me what the portfolio is teaching us”

Needs trends across projects: recurring operational failures, repeated capability gaps, technology adoption changes, and assumptions that appear stale. The product surfaces evidence; the executive remains the decision-maker.

Jobs To Be Done

- When I inherit a system, help me reconstruct why it looks this way so I do not undo a deliberate trade-off by accident.
- When I face a new decision, show me similar historical decisions and whether their constraints still match.
- When a past choice produced a later incident or success, connect the outcome back to the original decision with calibrated causal language.
- When multiple projects reveal the same pattern, consolidate that evidence into an organizational observation instead of making me rediscover it project by project.
- When organizational assumptions stop matching reality, flag the mismatch for human reconsideration without silently rewriting policy.

Core Use Cases

Use case	Input	Output
Historical rationale	“Why don’t we use Kafka?”	Decision timeline + original reasons + constraints + exceptions + sources
New decision support	New project requirements + question	Comparable historical decisions + constraint delta + decision brief
Decision Drift	Old decision + new evidence	Drift explanation + changed assumptions + affected projects + confidence
Downstream impact	Decision + later incident/postmortem	Decision → implementation → event → consequence chain with causal confidence
Organizational learning	Many projects over time	Consolidated observations / mental models with supporting evidence
Executive insight	Portfolio-level question	Evidence-linked trend statement, not autonomous strategy

Product Principles

- 1. Memory before generation. Retrieve organizational memory before allowing the answer model to reason about company-specific history.
- 2. Conditional, not absolute. Store the circumstances around a decision so “rejected” never automatically means “forbidden forever.”
- 3. Time is data. Preserve both when an event occurred and when the system learned it.
- 4. Evidence survives synthesis. Every high-impact claim must lead back to retained evidence and source artifacts.
- 5. Observations evolve; history remains inspectable. New evidence may refine an organizational belief but must not erase the underlying facts.
- 6. Human authority stays explicit. Drift, causal links, capability gaps, and recommendations are prompts for review, not automatic policy changes.
- 7. One memory engine. Do not introduce a second vector/graph memory stack merely to duplicate Hindsight.

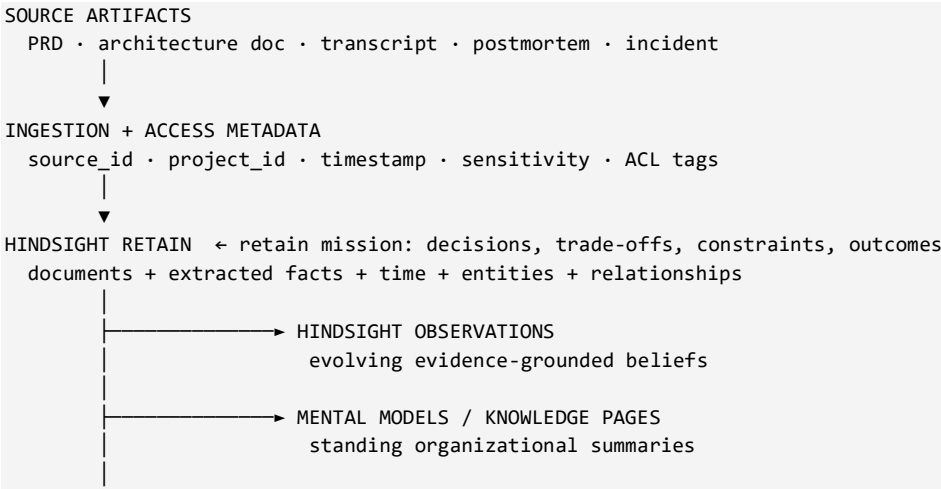
Competitive Differentiation

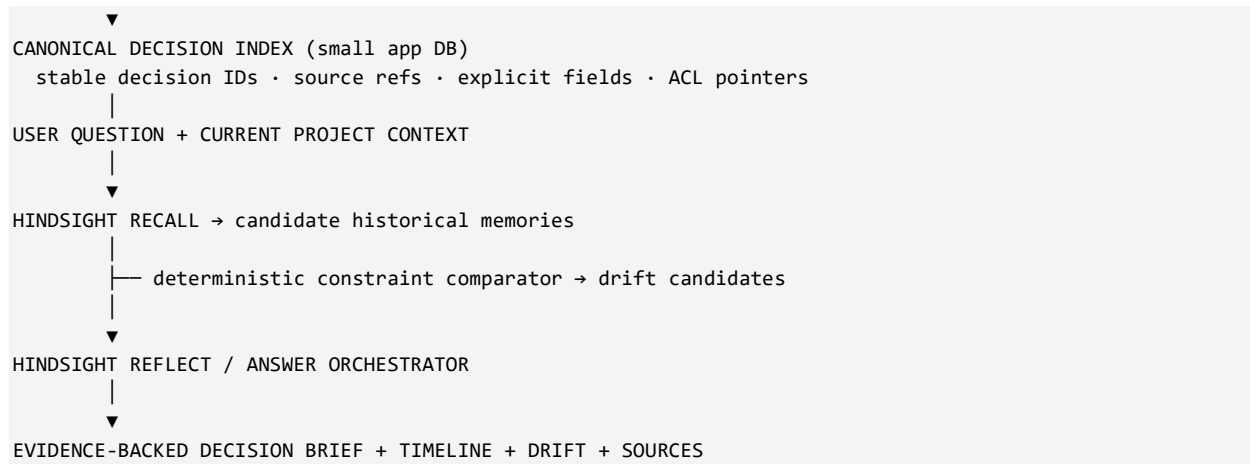
Category	Typical behavior	DecisionPrint difference
Generic RAG	Finds semantically similar chunks	Reconstructs time-bound decisions, reasons, constraints, and later outcomes
Enterprise search	Locates documents / passages	Builds an evolving organizational memory across documents and time
Traditional knowledge base	Human-maintained static pages	Observations and standing summaries can evolve as new evidence is retained
Vector database	Similarity retrieval primitive	Hindsight adds temporal, graph, keyword, semantic retrieval plus memory consolidation
Meeting assistant	Summarizes one meeting	Connects a meeting decision to older projects and later consequences
Generic copilot	Answers from current context	Reasons over persistent organizational history
Architecture Decision Records	Excellent explicit decision documents but manually created and often isolated	Extracts decision context from broader evidence and connects ADR-like decisions to outcomes and changing assumptions

Differentiator: DecisionPrint remembers not only WHAT the organization decided, but WHY, under WHICH constraints, WHAT happened afterward, and WHETHER those assumptions still hold.

Hindsight-Centric Memory Architecture

DecisionPrint uses one organization memory bank for the hackathon dataset, with strict tags/metadata for project, source class, sensitivity, and lifecycle. This choice enables cross-project learning — essential to the product — while allowing recall to be scoped to authorized projects. A production deployment can partition banks by tenant/business boundary where isolation requirements demand it.





Hindsight's official model aligns directly with this architecture: retain extracts structured facts and links entities; recall runs semantic, keyword, graph, and temporal retrieval; observations consolidate repeated evidence; and reflect can reason over observations, mental models, and raw facts. DecisionPrint therefore does not need a separate vector database or custom knowledge graph for the MVP.

Memory Model

Memory layers

Layer	Representation	Owner	Purpose
Raw source	Original document/transcript + source identity	Source store + Hindsight document association	Audit, exact evidence, reprocessing
Atomic memory facts	World / experience facts with time, context, tags, entities	Hindsight	What happened, who/what/when/why
Decision projection	Canonical Decision object pointing to memory/source IDs	Application DB	Deterministic UI, lifecycle, dedupe, explicit status
Observations	Consolidated evidence-grounded beliefs	Hindsight	Patterns, repeated exceptions, durable organizational learning
Mental models / pages	Standing synthesized answers	Hindsight	Fast living summaries such as architecture principles or recurring risks
Derived signals	Constraint delta, drift score, causal confidence	Application logic	Deterministic, inspectable product features

Representing concepts

Concept	Hindsight representation	Application representation
Decision	Retained fact(s) tagged decision + entities + occurred time	Decision ID, normalized title/status, source refs
Reason	Fact text preserving "why"; linked via shared source/entities	Structured reason field for display/extraction QA
Constraint	Retained fact tagged constraint; project/technology entities	Normalized key/value where extractable for comparison
Assumption	Retained fact tagged assumption	Assumption list + active/invalidated status
Alternative / rejection	Facts describing option and reason for rejection	Alternative rows linked to decision
Technology	Canonical entity + fact relationships	Technology label / filter
Project	Entity + project tag/metadata	Project record and ACL scope
Stakeholder	Entity when present in authorized source	Participant IDs/names only as needed; no performance scoring
Outcome / incident	Later retained event with temporal and entity links	Outcome record / incident link
Evidence / source	Document/chunk association on recalled fact	Source table with page/meeting timestamp if available
Confidence	Evidence/proof and retrieval information	Extraction confidence + inference confidence, separately displayed
Current status	Newer facts/observations over time	active / reconsidered / superseded / exception

Important implementation rule: Hindsight memory is the semantic/temporal truth substrate; the Decision object is a projection optimized for UI and deterministic comparisons. The app database must never become a shadow knowledge base containing all organizational memory.

Decision Model

Canonical Decision object

```
Decision {
  id: string
  project_id: string
  title: string
  decision_statement: string
  occurred_at: datetime
  participants: [entity_ref]
  context_summary: string
  constraints: [{name, value, unit?, source_memory_id}]
  assumptions: [{statement, status, source_memory_id}]
  alternatives: [{option, disposition, reason, source_memory_id}]
  selected_option: string
  reasons: [{statement, source_memory_id}]
  technologies: [entity_ref]
  source_refs: [{document_id, chunk_id?, locator?}]
  extraction_confidence: 0..1
  outcome_refs: [outcome_id]
  status: active | reconsidered | superseded | exception
  superseded_by: decision_id?
  related_decisions: [decision_id]
}
```

Application metadata: stable IDs, normalized fields, status, supersession pointers, source locators, and explicit confidence. Hindsight memory: the underlying decision facts, reasons, temporal context, entities/relationships, outcomes, observations, and learned organizational patterns.

Decision Lifecycle

```
PROPOSED / DISCUSSED
  ↓ evidence of commitment
DECIDED
  ↓ implementation / later events
OBSERVED OUTCOME
  ↓ new constraints or contradictory evidence
RECONSIDERATION CANDIDATE
  ├── no change → ACTIVE WITH NEW EVIDENCE
  ├── scoped exception → EXCEPTION
  └── changed decision → SUPERSEDED → NEW DECISION
```

Decisions are append-only historical events. A superseding decision does not overwrite the original memory. The application updates lifecycle metadata while Hindsight retains both historical and newer evidence, enabling temporal recall and inspection of evolution.

Decision Drift

Decision Drift identifies when the premises supporting an older decision appear materially different from current evidence. It is a review signal, not an automated reversal.

Drift triggers

- A historical constraint no longer holds.

- Repeated exceptions accumulate against a broad rule.
- A previously rejected technology increasingly appears in successful projects.
- The same decision is repeatedly reopened.
- Operational evidence contradicts an assumption that supported the original decision.

MVP drift algorithm

8. Recall historical decision memories relevant to the current question and entities.
9. Extract/normalize comparable constraints from the historical decision and current project context.
10. Classify each pair as same, changed, newly present, missing/unknown, or incomparable.
11. Compute an explainable drift score from material changed constraints, weighted by whether the historical reason explicitly depended on them.
12. Ask Hindsight reflect to synthesize the organizational context using recalled evidence, but require the UI to show the deterministic constraint delta beside the synthesis.
13. Label the result “reconsideration warranted” only when threshold conditions are met; never mutate the decision automatically.

```
drift_score = weighted_changed_reason_constraints / weighted_known_reason_constraints
```

Example:

```
consumer_count: 2 → 15      CHANGED (material)
replay_required: no → yes    CHANGED (material)
traffic: moderate → high     CHANGED (material)
team_size: small → unknown   UNKNOWN
```

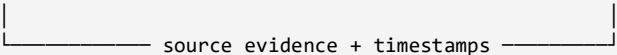
Signal: HIGH DRIFT · original premises materially differ

Causal / Downstream Impact Memory

DecisionPrint connects a decision to later outcomes only when evidence supports the link. The system must preserve calibrated causal language.

Label	Use when	Example wording
FACT	The source explicitly states the decision/event	“Backups were disabled in the deployment configuration.”
EXPLICIT CAUSAL LINK	A postmortem explicitly names the decision as a cause/contributor	“The postmortem identifies missing automated backups as a contributing factor.”
STRONG EVIDENCE	Multiple sources support a relationship but do not explicitly declare causation	“Evidence strongly associates the recovery delay with the backup decision.”
POSSIBLE CAUSAL LINK	Temporal/entity connection exists but causal support is incomplete	“This decision may be related; the available sources do not establish causation.”

Decision → Constraint → Implementation → Event → Incident → Consequence



For the MVP, causal chains are assembled from recalled facts sharing decision/project/technology entities plus temporal order, then classified by an LLM into the labels above. The classifier must return evidence IDs. If it cannot cite evidence, the link is not displayed as causal.

Functional Requirements

ID	Requirement	MVP priority
FR-01	Upload and ingest historical project documents and transcripts with project/date/source metadata.	P0
FR-02	Configure Hindsight retain mission to focus extraction on decisions, trade-offs, constraints, assumptions, alternatives, outcomes, incidents, and exceptions.	P0
FR-03	Extract canonical Decision objects from retained source material and link every field to evidence.	P0
FR-04	Retain current-project meeting transcripts and make new evidence available to later recall.	P0
FR-05	Ask natural-language questions scoped to authorized organizational/project memory.	P0
FR-06	Recall relevant historical decisions with temporal/entity context.	P0

FR-07	Compare old and current constraints and surface Decision Drift.	P0
FR-08	Generate a decision brief separating fact, observation, inference, and recommendation.	P0
FR-09	Display source evidence for every important claim.	P0
FR-10	Show decision timeline and supersession/evolution.	P0
FR-11	Ingest a later outcome/postmortem and connect it to the decision when supported.	P0 demo
FR-12	Show organizational observations/trends from multiple projects.	P1
FR-13	Executive dashboard with evidence-linked trends.	P1
FR-14	Manual correction/supersession of extracted Decision projection.	P1

Non-Functional Requirements

Area	Requirement
Traceability	100% of displayed historical claims must include at least one source/memory reference.
Latency	Target <5 s for recall-based brief; <12 s for deep reflect in demo environment.
Reliability	Failed consolidation/reflect must not destroy retained source or canonical decision records.
Security	No cross-project disclosure outside authorized scope; ACL filters applied before answer generation.
Explainability	Constraint delta and confidence visible without opening raw logs.
Auditability	Log query, authorized scope, recalled memory IDs, decision IDs, generated inference, and source references.
Data integrity	Corrections append/supersede; do not silently rewrite historical evidence.
Usability	A judge must understand the old decision, changed constraint, and drift signal in <20 seconds.

User Journeys

New employee journey

14. Ask “Why do we use REST instead of GraphQL in Billing?”
15. System recalls Billing and related API decisions, prioritizing authorized evidence.
16. Answer shows original decision, constraints, later exceptions, and source excerpts.
17. User opens timeline to see whether the organization’s position changed.

Project lead journey

18. Create/select current project and enter current constraints.
19. Ask “Should we use Kafka?”
20. DecisionPrint recalls prior Kafka decisions and observations.
21. Constraint comparator shows old vs new context.
22. Drift card explains which original premises changed.
23. Decision brief provides evidence, inference, confidence, and source links; lead decides.

Executive journey

24. Open Organizational Trends.
25. See “real-time processing demand appears in 6 of last 8 projects” with evidence count.
26. Open evidence panel to inspect supporting projects and constraints.
27. Treat “potential streaming capability gap” as a review prompt, not an automated hiring recommendation.

MVP Scope

Build one polished longitudinal decision loop. Everything else is secondary.

- One fictional company memory bank with 20–50 coherent historical projects and 5 current projects.
- Historical ingestion for text/Markdown/PDF-extracted text or prepared synthetic documents.
- Current meeting transcript ingestion.
- Decision extraction into canonical Decision objects with source references.
- Hindsight retain with timestamps, project/source tags, context, and retain mission.

- Hindsight recall for historical candidates and reflect for deeper synthesis.
- Automatic observations allowed to consolidate organizational patterns; 2–4 curated mental models for standing summaries.
- Deterministic old-vs-current constraint comparison and Decision Drift card.
- Evidence-backed decision brief.
- Decision timeline showing evolution/supersession.
- One downstream incident/postmortem chain demonstrating outcome memory.
- Streamlit UI optimized for the killer demo.

Explicitly Out of Scope

- Autonomous architecture approval or policy mutation.
- Autonomous hiring, firing, employee ranking, or performance surveillance.
- Production Slack/Teams/Jira/GitHub/Confluence integrations.
- Real-time meeting bot.
- General-purpose enterprise search replacement.
- Full causal inference engine.
- Decision impact prediction.
- Fine-grained production IAM integration beyond demo RBAC/ACL tags.
- A second vector database or graph database duplicating Hindsight memory.

Technical Architecture

Component	Responsibility	Suggested MVP technology
Streamlit UI	Upload, project context, ask, timeline, drift, evidence	Python + Streamlit
Application service	Orchestration, auth scope, Decision CRUD, comparison, audit	Python/FastAPI or in-process Streamlit service
Hindsight client	retain / recall / reflect / observations / mental models	Official Hindsight Python client
Extraction model	Structured Decision projection from source/retained evidence	LLM with JSON schema / Pydantic validation
App DB	Projects, canonical decisions, source ACLs, supersession, audit	SQLite for hackathon
Source store	Original synthetic files and stable source IDs	Local filesystem
Analytics	Simple counts/trends over Decision projection + Hindsight observations	Python/pandas
LLM provider	Extraction and answer orchestration where Hindsight config requires	Hackathon-supported model/provider

Architecture rule: SQLite is not a memory system. It exists for deterministic application state and UI joins. Organizational facts and longitudinal learning live in Hindsight.

Data Architecture

Application tables

Table	Minimal fields
projects	project_id, name, status, start_date, end_date, acl_scope
sources	source_id, project_id, type, title, occurred_at, path, sensitivity, hindsight_document_id
decisions	decision_id, project_id, statement, occurred_at, status, confidence, superseded_by
decision_constraints	decision_id, key, value, normalized_value, evidence_memory_id
decision_evidence	decision_id, source_id, memory_id, chunk_id, locator, evidence_role
outcomes	outcome_id, project_id, occurred_at, statement, causal_label, confidence
audit_events	query_id, user_role, scope, recalled_memory_ids, output_hash, timestamp

Tag / metadata convention

```
tags: [
  "org:acme-labs",
  "project:alpha",
  "source:architecture-decision",
  "domain:streaming",
  "visibility:engineering"
]
metadata: {
  source_id: "SRC-ALPHA-014",
  project_id: "alpha",
  source_type: "meeting_transcript",
  sensitivity: "internal"
}
```

Hindsight Integration

Bank strategy

MVP: one organization bank to enable cross-project recall, with project and visibility tags. Production: one bank per tenant or hard isolation boundary; use tags only inside a boundary where cross-project learning is authorized. Hindsight banks are isolated memory stores, so bank boundaries should align with real disclosure boundaries rather than arbitrary UI organization.

Retain

Every source is retained with source context, project tags/metadata, and an event timestamp where known. The bank retain mission should explicitly prioritize technical/product decisions, alternatives, constraints, assumptions, reasons, outcomes, incidents, exceptions, and downstream consequences while ignoring meeting logistics.

```
retain_mission:
"Extract durable organizational decision memory. Preserve what was decided, why,
constraints and assumptions at the time, alternatives considered/rejected, relevant
technologies and stakeholders, dates, dependencies, outcomes, incidents, exceptions,
and explicit causal claims. Ignore greetings, scheduling, and low-value chatter."
```

Use verbose extraction for high-value decision/postmortem sources if hackathon latency permits; use concise for bulk background material. Store stable source IDs so recalled facts can be traced to their document/chunk associations.

Recall

For decision questions, recall across world, experience, and observation memories, preferring observations where supported while retaining access to unconsolidated recent facts. Apply project/visibility tag filters based on authorization. Request entity information and use temporal query context for “at the time” questions.

Reflect

Use reflect for questions requiring synthesis across multiple memories: recurring risks, why organizational behavior changed, whether evidence suggests a prior assumption is stale, or what prior outcomes matter. Do not use reflect as a substitute for the deterministic constraint comparator. The answer layer should receive the comparator output as additional context and return a structured brief.

Observations

Observations are crucial to “living memory.” They consolidate repeated evidence into deduplicated beliefs and are refined as supporting or contradictory evidence arrives. DecisionPrint uses them for patterns such as “small teams repeatedly avoid operationally heavy infrastructure” or “streaming requirements are increasingly common.” The UI labels these as OBSERVATION, not historical fact.

Mental models / knowledge pages

Create only a few high-value standing summaries for the demo, for example “Current architecture principles,” “Recurring operational risks,” and “Technology adoption patterns.” Mental models are appropriate because they are saved reflect responses that can be refreshed as memory changes. Knowledge pages are useful if the team wants a wiki-like view, but they are optional to the killer loop.

Memory history and source association

Use memory/document IDs and observation history where available to show that an organizational belief has accumulated or changed evidence. The UI does not need to expose Hindsight internals everywhere; it should expose the evidence count, supporting sources, and a simple “memory evolved” timeline.

LLM Architecture

Stage	Model task	Guardrail
Decision extraction	Convert source into canonical decision JSON	Schema validation; every field requires evidence ref or null
Constraint normalization	Map comparable quantities/booleans/categories	Deterministic normalization first; LLM only for semantic equivalence
Causal classification	Classify explicit / strong / possible / none	Must return evidence IDs and rationale
Decision brief	Synthesize recalled memory + current context + delta	Structured labels: FACT / OBSERVATION / INFERENCE / RECOMMENDATION
Executive insight	Summarize repeated evidence	Minimum evidence threshold; never imply prediction or autonomous action

```
DecisionBrief {
  answer_summary
  historical_decisions[]
  historical_constraints[]
  current_constraints[]
  constraint_differences[]
  observations[]
  inferences[]
  drift_signal: none|low|medium|high
  reconsideration_warranted: boolean
  recommendation_note
  confidence
  sources[]
}
```

UI/UX

Navigation

View	Purpose
Organizational Memory Overview	Memory counts, projects, decisions, observations, recent evolution
Current Projects	Current constraints, recent meetings, unresolved decisions
Decision Explorer	Filter decisions by project/technology/date/status; inspect rationale
Ask DecisionPrint	Primary question experience and decision brief
Decision Timeline	Original decision → exceptions → outcomes → supersession
Decision Drift	Cards for assumptions/constraints that no longer match
Evidence Panel	Source excerpts, dates, project, fact/observation labels
Project Impact Graph	Small graph of decision → outcome links for selected decision
Organizational Trends	Evidence-backed recurring patterns and capability signals

Primary Ask screen

Ask DecisionPrint: Should Project Nova use Kafka?			
CURRENT CONTEXT			
15 consumers · replay required · high volume · async workflows			
HISTORICAL MEMORY		CONSTRAINT DELTA	
Project Alpha, 2025	consumers	2 → 15	CHANGED
“Do not use Kafka”	replay	no → yes	CHANGED
Reason: complexity	traffic	moderate → high	CHANGED
⚠ DECISION DRIFT: HIGH			
Original premises materially differ. Reconsideration warranted.			
FACTS · OBSERVATIONS · INFERENCE · CONFIDENCE · SOURCES			

Security & Privacy

- Confidential information: store only in authorized tenant/bank boundaries; encrypt storage/transport in production.
- Project isolation: authorization scope is resolved before recall; tag filters are derived server-side, never trusted from client input.
- Sensitive transcripts: classify source sensitivity and allow source-level exclusion from broad executive views.
- Employee privacy: retain organizational decisions and work context, not private life or performance scoring. Do not create “employee quality” observations.
- Client information: use project/client visibility tags and redact synthetic demo data from real client identifiers.
- Accidental disclosure: answer generator receives only authorized recalled evidence.
- Deletion/correction: delete or reprocess source documents where supported; mark canonical decisions corrected/superseded; preserve audit records consistent with policy.
- Stale memory: show dates, status, and drift; do not present old observations as timeless truth.
- Prompt injection in documents: treat source content as evidence, not instructions; extraction prompts explicitly ignore embedded commands.
- Hallucination/attribution: no historical claim without evidence reference; low-confidence extraction is queued for review.

Trust & Explainability

Every important answer uses a fixed epistemic structure so users can see what the system knows versus what it inferred.

```
ANSWER
↓
HISTORICAL DECISIONS [FACT]
↓
EVIDENCE + DATES + SOURCES
↓
ORIGINAL CONSTRAINTS
↓
CURRENT DIFFERENCES
↓
ORGANIZATIONAL PATTERNS [OBSERVATION]
↓
SYSTEM REASONING [INFERENCE]
↓
POSSIBLE ACTION [RECOMMENDATION – human decides]
↓
CONFIDENCE + SOURCE LINKS
```

Confidence is multi-dimensional: extraction confidence (did we parse the decision correctly?), retrieval confidence/relevance, drift confidence (how many reason-linked constraints are comparable?), and causal confidence. Do not collapse these into one magical percentage unless the UI explains its meaning.

Evaluation Framework

Metric	Definition	Target for demo
Decision extraction accuracy	Correct decision statement + project + date	≥90% on curated test set
Reason retrieval accuracy	Correct historical reason in top evidence	≥90%
Constraint matching accuracy	Correctly aligns old/current comparable constraints	≥85%
Historical decision retrieval	Relevant prior decision in top 3	≥90%
Decision Drift F1	Correct drift/no-drift on labeled scenarios	≥0.85
Source attribution accuracy	Claims point to supporting source	≥95%
Causal-chain precision	Displayed causal labels supported by evidence	≥90% precision
Contradiction detection	Flags deliberate conflicting evidence	≥80%
Hallucination rate	Unsupported historical claims	<5%; goal 0% in killer demo
Latency	Question → rendered brief	<5 s recall path; <12 s reflect path

Baseline Comparison

Run the same 15–25 labeled decision questions through two systems: (A) generic RAG over source chunks using semantic retrieval and (B) DecisionPrint + Hindsight with longitudinal memory, observations, temporal/entity retrieval, and drift comparison. Keep the answer model and source corpus constant where possible.

Test	Generic RAG expectation	DecisionPrint + Hindsight hypothesis
"Why was Kafka rejected?"	Often retrieves the rejection passage	Retrieves rejection plus reason, time, entities, related evidence
"Do those reasons still apply?"	Requires current context to coincidentally retrieve matching chunks	Explicit old-vs-current constraint comparison backed by temporal memory
"Has our position changed?"	Hard to synthesize reliably across scattered dates	Observations + temporal recall + decision timeline expose evolution
"What happened afterward?"	May retrieve postmortem if semantically similar	Entity/temporal links and retained outcomes support decision→consequence chain
"What pattern exists across projects?"	One-shot synthesis over retrieved subset	Observations/mental models provide accumulated organizational learning

Primary experiment: measure not only retrieval hit rate but “contextual decision correctness” — whether the answer correctly identifies that a historical decision was conditional and whether current constraints materially differ.

Demo Scenario

Fictional company: Northstar Systems

Northstar builds B2B platforms. In 2025, Project Alpha rejected Kafka because it had two consumers, moderate traffic, no replay requirement, and a small operations team. The decision was successful under those conditions. Later projects gradually introduce more asynchronous workloads. In 2026, Project Nova requires 15 consumers, replay, high event volume, and asynchronous processing.

A later synthetic postmortem adds a second learning chain: Project Cedar skipped automated database backups to reduce infrastructure cost. A database failure caused prolonged recovery, and the postmortem explicitly identifies missing backups as a contributing factor. This demonstrates that DecisionPrint remembers downstream consequences, not merely decisions.

Demo Script

Time	Action	What judge sees / why Hindsight matters
0–15s	Open Organizational Memory. Search	Historical memory: Project Alpha — “Do not introduce Kafka.” Source/date

	"Kafka."	visible.
15–30s	Open decision.	Reason + old constraints: 2 consumers, no replay, moderate traffic. Not a timeless rule.
30–45s	Ingest Nova meeting transcript or reveal it as newly retained evidence.	Current evidence enters the same living memory; project context shows 15 consumers + replay + high volume.
45–65s	Ask "Should Nova use Kafka?"	Recall brings Alpha; comparator highlights changed premises; Decision Drift = HIGH; brief says reconsideration warranted, not "Kafka is automatically correct."
65–78s	Open Memory Evolution / observation.	Organizational view now shows increasing streaming requirements / exceptions as evolving evidence.
78–90s	Open Cedar incident chain.	Decision → implementation → incident → consequence, labeled EXPLICIT CAUSAL LINK with postmortem source.

Judge takeaway in one glance: Hindsight changes the answer because the product remembers the original context and later evidence across time. Remove Hindsight and the defining behavior collapses into document retrieval.

Synthetic Dataset Design

The dataset must be a connected organizational history, not a pile of random files. Generate projects in families so constraints and decisions evolve.

Project family	Historical arc	Technologies / themes
Eventing & integration	2 consumers/no replay → 5 consumers → 15 consumers/replay	Kafka, RabbitMQ, webhooks, queues
Caching & scale	single instance → 4 instances → 20 instances	Redis, in-process cache
API design	3 known clients → partner ecosystem → mobile/web/public API	REST, GraphQL, versioning
Data reliability	cheap MVP → production SLA → incident-driven hardening	PostgreSQL, backups, replicas
Observability	manual debugging → distributed services	logging, tracing, metrics
Deployment	single service → multi-service / regional needs	containers, CI/CD, orchestration

Dataset size

- 5 current projects.
- 30 historical projects (enough breadth without slowing the hackathon).
- 60–80 explicit decisions.
- 24 meeting transcripts.
- 8 architecture documents / ADR-like files.
- 6 retrospectives.
- 5 postmortems / incidents.
- 3–5 repeated decision themes with deliberate constraint evolution.
- At least 4 superseded/reconsidered decisions.
- At least 3 recurring exceptions.
- At least 2 explicit downstream consequence chains.

Required connected chains

28. Kafka chain: reject under low complexity → increasing streaming demand → Nova drift.
29. Redis chain: reject for single instance → multi-instance scaling → shared cache becomes relevant.
30. GraphQL chain: reject for three clients → later partner ecosystem creates new client diversity; perhaps still rejected for other reasons, demonstrating drift does not dictate outcome.
31. Backup chain: cost-saving decision → implementation → failure → postmortem explicit contributing factor.
32. Tracing chain: initially "logs are sufficient" → distributed services → repeated debugging delays → organizational observation that tracing becomes increasingly necessary.

Metrics

The hackathon dashboard can show evaluation results on a held-out set of synthetic questions. The most persuasive chart is paired accuracy: generic RAG versus DecisionPrint + Hindsight on reason retrieval, temporal/context match, and drift detection. Avoid claiming benchmark significance from a tiny synthetic set; present it as an internal demo evaluation.

Hackathon Judging Strategy

Criterion	Weight	How DecisionPrint earns it
Innovation	30%	Decision memory + constraint-aware drift + downstream consequence chains; not another chat-with-docs app.
Hindsight Memory	25%	Core retain→observations→recall/reflect loop; temporal/entity retrieval; living mental models; demo visibly depends on evolving memory.
Technical Implementation	20%	Canonical decision projection, evidence mapping, deterministic constraint comparator, ACL-aware retrieval, structured answer schema.
UX	15%	Old-vs-new constraint delta, drift card, timeline, evidence panel, epistemic labels make complex memory understandable quickly.
Real-world Impact	10%	Reduces repeated mistakes, onboarding loss, stale architectural dogma, and fragmented institutional knowledge.

How to make Hindsight impossible to overlook

33. Put “Memory learned from 30 historical projects” on the opening screen with live counts from the bank/application projection.
34. During the demo, show the old Kafka decision before adding/revealing Nova evidence.
35. Retain the Nova transcript and visibly show “memory updated.”
36. Ask the same conceptual question after new evidence and show changed reasoning / drift, not merely a different retrieved chunk.
37. Open the evidence/observation view to show accumulated support and temporal evolution.
38. Explicitly state which product behaviors come from Hindsight: long-term retained facts, entities/relationships, temporal recall, observations, reflect, mental models.

Implementation Plan

Phase	Hours	Deliverable
1. Memory spike	0–3	Run Hindsight locally/cloud; configure bank; retain 3 sample docs; validate recall/reflect and source IDs.
2. Dataset + schemas	3–7	Generate connected synthetic corpus; Pydantic Decision schema; project/source metadata.
3. Ingestion	7–12	File/transcript ingestion → retain; canonical decision extraction; SQLite linkage.
4. Query loop	12–18	Recall + structured decision brief + evidence panel.
5. Drift	18–23	Constraint normalization/comparator + drift explanation.
6. Outcome chain	23–27	Postmortem ingestion + causal labels + impact graph.
7. UX polish	27–34	Streamlit screens, timeline, memory overview, demo transitions.
8. Evaluation	34–38	Labeled questions, generic-RAG baseline, metric table.
9. Demo hardening	38–44	Seed deterministic dataset, cache expensive calls, failure states, 90-second rehearsal.
10. Buffer	44–48	Bug fixes, screenshots, pitch, README.

Team Responsibilities

Role	Primary ownership
Memory / backend engineer	Hindsight setup, retain/recall/reflect, tags, observations/mental models, source traceability
AI / extraction engineer	Decision schema, extraction, normalization, causal classifier, structured brief
Product / frontend engineer	Streamlit UX, timeline, drift visualization, evidence panel, demo flow
Data / evaluation owner	Synthetic corpus, connected temporal chains, labels, baseline comparison, metrics
All team members	Demo rehearsal, edge cases, judge narrative, integration testing

For a 2–3 person team, merge roles: backend+AI, frontend+product, data+evaluation. Do not split into microservices or over-engineer deployment.

Risks

Risk	Why it matters
Extraction misses a decision	Breaks the visible memory chain.
Observation consolidation timing	New evidence may not instantly appear as a consolidated observation.
Reflect is slow or variable	Demo latency / nondeterminism.
Synthetic data feels fake	Judges may discount real-world impact.
Too much UI	Core memory story gets diluted.
ACL filtering error	Could expose confidential cross-project content.
Causal overclaim	Damages trust.
Hindsight becomes hidden plumbing	Loses the 25% memory criterion even if technically used.

Mitigations

Risk	Mitigation
Extraction miss	Seed critical demo decisions; run ingestion QA and show low-confidence review state.
Observation timing	Use raw facts + prefer-observations recall; pre-consolidate demo corpus; do not depend on immediate observation creation for the critical path.
Reflect variability	Cache demo-safe results after memory state is seeded; retain live interaction for at least one step; use structured schema.
Synthetic credibility	Write realistic documents with consistent dates, stakeholders, constraints, and later references; include messy but coherent meeting language.
UI scope	Polish Ask + Drift + Evidence + Timeline first; executive screens are secondary.
ACL	Server-side scope resolution; test unauthorized project query.
Causality	Require evidence IDs and calibrated labels; default to “possible” or no link.
Hindsight visibility	Add a Memory Trace drawer showing retained/recalled/observation evidence in human terms, not raw API logs.

Future Roadmap

- Direct meeting ingestion and meeting agents.
- Slack / Teams integration.
- GitHub, Jira, and Confluence connectors.
- Continuous project monitoring and automatic decision detection.
- Automated postmortem ingestion.
- Organization-wide knowledge pages with review/approval workflows.
- Richer executive intelligence and capability planning.
- Human-reviewed hiring/capability recommendations based on repeated project evidence.
- Service opportunity discovery from recurring customer requirements.
- Interactive decision graph visualization.
- Decision impact prediction only after sufficient validated outcome data.
- Policy workflows where authorized humans approve organizational principles derived from observations.

One-Sentence Pitch

DecisionPrint is a Hindsight-powered organizational memory that remembers what your company decided, why it decided it, the constraints that made the decision sensible, what happened afterward, and when those assumptions stop being true.

30-Second Pitch

Companies remember decisions but forget the reasoning around them. DecisionPrint turns PRDs, meetings, architecture docs, and postmortems into a living decision memory powered by Hindsight. When a team asks, “Should we use Kafka?”, it does not just retrieve an old note saying Kafka was rejected. It remembers that Kafka was rejected when there were only two consumers and no replay requirement, compares that with today’s 15 consumers and replay needs, flags Decision Drift, and shows the evidence. It also remembers what happened after past decisions, so the organization learns instead of repeatedly starting from zero.

2-Minute Pitch

Every organization accumulates hundreds of technical and product decisions. The final choice might be written down, but the reasoning is fragmented across meetings, PRDs, architecture docs, and postmortems. Months later, “we decided not to use Kafka” becomes folklore — the original constraints disappear.

DecisionPrint is an organizational decision-memory system built around Hindsight. We retain historical and current project evidence into a long-term memory bank. Hindsight preserves facts, time, entities, relationships, and evolving observations. When a new project lead asks a question, DecisionPrint recalls comparable historical decisions and shows the conditions under which they were made.

Our killer example is Kafka. Project Alpha rejected Kafka because it had two consumers, moderate traffic, and no replay. Six months later Project Nova has 15 consumers, high volume, replay, and asynchronous workflows. DecisionPrint does not say the old decision was wrong and it does not automatically choose Kafka. It shows that the assumptions behind the old decision have drifted and that reconsideration is warranted.

Then we go one step further. If a later postmortem says a past decision contributed to an incident, DecisionPrint connects the decision to the downstream consequence with explicit causal confidence. Over time, Hindsight observations and mental models turn individual project evidence into living organizational learning. This is not chat with documents. It is memory for how an organization reasons.

Final Product Definition

DecisionPrint is a Hindsight-native organizational decision-memory agent that continuously converts project evidence into temporal, evidence-backed memory of decisions, reasons, constraints, alternatives, outcomes, and evolving organizational beliefs; retrieves that memory when new decisions arise; detects when historical premises no longer match current reality; and presents the result as a traceable decision brief while leaving final judgment to humans.

Implementation Notes — Verified Hindsight Capabilities

The architecture above is grounded in current official Hindsight documentation and repository documentation reviewed in September 2026. The implementation should re-check exact client method signatures against the version installed during the hackathon.

Capability used by DecisionPrint	Verified behavior / relevance
Memory banks	Isolated containers for memories, documents, entities, relationships, and related bank state; useful for tenant/hard-boundary isolation.
Retain	Processes content into facts, temporal data, entities and relationships; supports context, timestamps, tags/metadata, extraction missions/modes, and source/document association.
Recall	Runs semantic, keyword/BM25, graph, and temporal retrieval; can filter memory types/tags and return entities/document/chunk associations.
Memory types	World and experience facts plus observations; observations are consolidated evidence-grounded beliefs.
Observations	Created/refined in background from multiple facts; preserve supporting evidence and evolve rather than simply overwrite.
Reflect	Deeper reasoning over memory for synthesis questions; can use mental models, observations, and raw facts hierarchically.
Mental models	Saved/pre-computed reflect responses for standing questions; suitable for organizational principles/trends that refresh over time.

Knowledge pages	Wiki-like living documents built on mental-model mechanics; optional for MVP.
Memory history	CLI/API capabilities expose observation history for a memory unit; useful for evolution/audit views.

Official references

- **Hindsight repository / overview:** <https://github.com/vectorize-io/hindsight>
- **Quickstart — retain / recall / reflect:** <https://github.com/vectorize-io/hindsight/blob/main/skills/hindsight-docs/references/developer/api/quickstart.md>
- **Retain documentation:** <https://github.com/vectorize-io/hindsight/blob/main/skills/hindsight-docs/references/developer/retain.md>
- **Recall documentation:** <https://github.com/vectorize-io/hindsight/blob/main/skills/hindsight-docs/references/developer/api/recall.md>
- **Memory banks:** <https://github.com/vectorize-io/hindsight/blob/main/skills/hindsight-docs/references/developer/api/memory-banks.md>
- **Mental models:** <https://github.com/vectorize-io/hindsight/blob/main/skills/hindsight-docs/references/developer/api/mental-models.md>
- **Best practices / taxonomy:** <https://github.com/vectorize-io/hindsight/blob/main/skills/hindsight-docs/references/best-practices.md>